Federal Reserve Bank *of* Atlanta

Labor Market Sliders

This tool illustrates the dynamic relationship between employment growth, the unemployment rate, and GDP growth. Users can explore "what if" questions about labor market variables and GDP.

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Have you ever wondered what would happen to employment and gross domestic product (GDP) growth if the unemployment rate hit 3 percent? If productivity hit 2 percent over the next year? Or what would happen to the unemployment rate over the next three years if the labor force participation rate rose sharply, but employment growth stayed constant? This tool — the labor market sliders — is designed to answer a variety of these "what if" questions over different time horizons. By using the mathematical relationships behind the data (for example, GDP's relationship to employment growth and productivity), the sliders provide convenient insight into the connections among five important macroeconomic indicators: GDP, labor productivity, employment, unemployment, and the labor force participation rate. chart icon to the right of each slider to see the history of each data series.

Unemployment Rate

(percent of people in the labor force unemployed)

Reset

Employment

(average monthly change in number of jobs)

Reset

GDP Growth

(average annualized growth in total output)

Reset

Labor Productivity

(average annual growth in GDP per worker)

Reset

Labor Force Participation Rate

(percent of people in population either employed or unemployed)

Reset

Months

Reset

Reset All

1. Why does the Federal Reserve care about the labor market to begin with?

Section 2A of the Federal Reserve Act states, "The Board of Governors of the Federal Reserve System and the Federal Open Market Committee shall maintain long-run growth of the monetary and credit aggregates commensurate with the economy's long run potential to increase production, so as to promote effectively the goals of maximum employment, stable prices, and moderate long-term interest rates." This part of the Federal Reserve Act is often referred to as the Fed's "dual mandate." Basically, it states that the Federal Reserve's monetary policy has the goals of stable prices and maximum employment. The unemployment rate is the most popular statistic that measures the degree to which the Federal Reserve has achieved the goal of maximum employment.

2. Can I use the Labor Market Sliders as a forecasting tool?

No. The Labor Market Sliders tool is an illustration of the mathematical relationship between employment growth, the unemployment rate, and GDP growth, over a given time period, and for specific values of productivity growth and labor force participation. It is not a forecasting tool, but,

rather, is driven by simple algebraic formulas that are described in greater detail in the **Behind the Scenes** tab. To explore GDP forecasting, see another Atlanta Fed tool called GDPNow.

3. What are the sources of the numbers behind the Labor Market Sliders?

The current labor market statistics reported each month by the U.S. Bureau of Labor Statistics (BLS) are used to support the LMS. These statistics are derived from two surveys, the Household Survey and the Establishment, or Payroll, Survey . (For definitions of terms and concepts, consult the official BLS Handbook of Methods .) Gross domestic product (GDP) is obtained from the Bureau of Economic Analysis (BEA). Worker productivity (or output per worker) is constructed by dividing the current GDP by the current value of CES employment. See the **Behind the Scenes** tab for more details. Of course, these sources supply the underlying statistics; the user can change the values and underlying time frame to see how the statistics relate to each other algebraically.

4. Why are employment changes much larger when I change labor force participation when my time frame is one month versus 12 months?

The employment change reflected in slider 2 is an average monthly result of the change in the labor force participation rate (LFPR) over the number of months reflected in the last slider. If the economy is only given one month, for example, to absorb an increase in the LFPR without changing the unemployment rate, the change in employment will have to be much larger than if that change in the LFPR is spread over a greater number of months.

5. The unemployment rate (UR) is a concept that comes from the Household Survey. How do you map changes in the UR to monthly changes in the Payroll (Establishment) Survey?

Since the headline jobs number refers to the payroll employment from the Establishment Survey and the unemployment rate comes from the Household Survey, the LMS makes a simple transformation, based on the average ratio of establishment/household employment over the past 12 months in order to translate changes in the unemployment rate to changes in payroll employment and vice versa.

6. How do the Payroll and Household Surveys differ in their concept of employment?

The Payroll and Household Surveys both claim to estimate the number of jobs in the economy. However, they are different for several reasons. The Establishment Survey estimates the number of jobs for which a paycheck was written in the United States during a particular pay period, whereas the Household Survey is a measure of employment *activity*. For example, if one person holds two jobs, the Establishment Survey will count the two jobs, but the Household Survey counts one person employed. In addition—and the main source of the difference—the scope of the Household Survey is broader than the Establishment Survey. The Household Survey includes the self-employed, unpaid family workers, agricultural workers, and private household workers—all of these workers are excluded from the Establishment Survey.

7. Why does the GDP growth reported on the Labor Market Sliders (slider 3) differ from that reported this quarter by the BEA?

Every quarter, the BEA reports the annual rate of the change in GDP from one quarter to the next. The LMS take a longer-term view and report the year-over-year difference in the current quarter estimate of the annual GDP (seasonally adjusted annual rate). From one quarter to the next, these figures can be quite different but will, on average over the year, converge to show similar trends.

However, even more importantly, the GDP reported by the BEA derives from a complicated macroeconomic accounting model that takes into account all activities in the economy. The LMS reflect the growth in GDP that is algebraically consistent with the unemployment rate reflected in slider 1, the average monthly employment growth reflected in slider 2, current average annual productivity growth reflected in slider 4, the LFPR reflected in slider 5, and the chosen time frame. It does not result from any specific economic model. See the **Behind the Scenes** tab for more details.

8. How does the productivity growth shown here differ from that reported by the Bureau of Labor Statistics?

While the concept of productivity (output per worker) is similar, the data used by the LMS to construct productivity differ from the data used to calculate productivity reported by the BLS . The first difference is that the LMS uses a concept of output per worker rather than output per worker hour. This difference is trivial when considering growth in productivity. As its measure of output, the BLS uses output in the nonfarm business sector rather than gross domestic product (GDP) for the whole economy. Correspondingly, the BLS uses nonfarm business sector employment rather than Payroll Survey employment and hours. The nonfarm business sector is a subset of the domestic economy and excludes the economic activities of the following: general government, private households, nonprofit organizations serving individuals, and farms. The nonfarm business sector accounted for about 77 percent of the value of gross domestic product in 2000.

9. How does the GDP/employment relationship reflected here differ from Okun's Law?

A statistical equation often used to describe the relationship between the unemployment rate and GDP growth is called Okun's Law . Far from being a law, it is more a rule of thumb . It is supposed to predict how much the unemployment rate will fall given a certain level of GDP growth. The prediction is based on an assumed linear relationship that does not necessarily conform to the rules of linearity. Because Okun's Law lacks theoretical foundation or consistent historical fortitude, the LMS does not apply any version of Okun's Law in the calculation of GDP growth that derives from the calculated additional employment.

10. Why does the average monthly change in employment reported on the Labor Market Sliders (slider 2) differ from the average monthly change in employment reflected in the pop-out graph?

The employment change reported on slider 2 reflects the change in employment that is algebraically consistent with the unemployment rate reflected in slider 1, the LFPR reflected in slider 5, and the chosen time frame. The change in payroll employment reflected in the pop-out graph is an average of the actual change in jobs over the previous 12 months, as reported by the BLS.

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Behind the Scenes of the Labor Market Sliders

The number of jobs an economy supports, the number of people who want jobs but don't have one, the number of people willing to work, how much output each worker can produce...these are all factors that define the state of the labor market and economic growth. The purpose of this page is to explain what is going on behind the scenes to produce the Labor Market Sliders (LMS).

The data used for the LMS all come from the U.S. Bureau of Labor Statistics (BLS) and the Bureau of Economic Analysis (BEA). Some data are updated monthly (the number of jobs created, the labor force participation rate, and the unemployment rate), and some are updated quarterly (growth in the gross domestic product, or GDP, and, hence, productivity growth).

Unemployment rate and payroll employment

The first two sliders reflect the unemployment rate and the average monthly change in payroll employment. These are the two statistics superstars reported by the BLS on the first Friday of every month . The payroll employment numbers and the unemployment rate are estimated by the BLS from responses to two separate surveys.

Through the Current Employment Statistics (CES) program, the BLS surveys approximately 141,000 nonfarm businesses, covering about 486,000 work sites, asking employers about employment, hours, and earnings of their workers. This survey of establishments is commonly referred to as the Payroll Survey or the Establishment Survey.

The total employment number that is reported from this survey reflects an estimate of the number of paychecks written in the United States for work performed during the pay period that includes the 12th day of the month. This count is considered to give the most accurate count of the total number of jobs in the United States at a given point in time.

The BLS also surveys about 60,000 households every month to obtain estimates of employment and nonemployment activity, total income, and demographics of the population of the United States (those 16 years and older). The reference period for activity is the same week as the Establishment Survey: the week that includes the 12th day of the month. The Household Survey is officially called the Current Population Survey (CPS).

From responses to the Household Survey, the BLS calculates the total labor force, the labor force participation rate, and the unemployment rate. While the user should consult the official BLS Handbook of Methods for complete definitions of these terms, here are some basic concepts and formulas:

Employed (E): A person who has a job for pay, is temporarily absent from work, or works unpaid in a family business.
Unemployed (U): A person who does not have a job, is available to work, and has looked for work during the previous four weeks.
Labor Force (LF): The sum of everyone who is employed and unemployed (restricted to noninstitutionalized civilians).
Labor Force Participation Rate (LFPR): The labor force divided by the population (POP), *LFPR = LF/POP*.
Unemployment Rate (UR): The number unemployed divided by the labor force, *UR = U/LF*.

GDP growth and productivity

The third slider reflects average annual growth in GDP. The LMS appeals to a very basic accounting relationship to calculate the growth in gross domestic product as a function of employment change. The relationship between GDP growth and employment depends on productivity, or, rather, the amount of output being produced by each worker.

The average annual growth in GDP expected over *t* number of months, given the employment growth reflected by slider two and assumed growth in productivity, is calculated using the following formula:

$$\%\Delta GDP = (12/t) * \{[(1+\%\Delta P)*P1*(\Delta E*t+E1)]-GDP1\}/GDP1$$

where $\%\Delta P$ is the average annual productivity growth (reflected in slider four), ΔE is the average monthly change in payroll employment over *t* months (reflected in slider two), *E1* is the current period level of payroll employment, *GDP1* is the current period level of GDP, and *P1* is the current period level of productivity [*GDP*1/*E1*]. Note that GDP is reported quarterly. Therefore, average annual GDP growth and annual productivity growth reported in the LMS will reflect the latest vintage of the most recent quarter for which GDP figures are reported.

Also note that the average annual growth in GDP reported for the current quarter is the year-over-year percentage change in the real value of the seasonally adjusted annual rate of quarterly GDP reported by the Bureau of Economic Analysis .

While the concept of productivity (output per worker) is similar, the data used by the LMS to construct productivity differ from the data used to calculate productivity reported by the BLS . The first difference is that the LMS uses a concept of output per worker instead of output per worker hour. This difference is trivial when considering growth in productivity. As its measure of output, the BLS uses output in the nonfarm business sector , rather than GDP for the whole economy. Correspondingly, the BLS uses nonfarm business sector employment rather than payroll survey employment and hours. The nonfarm business sector is a subset of the domestic economy and excludes the economic activities of the following: general government, private households, nonprofit organizations serving individuals, and farms. The nonfarm business sector accounted for about 77 percent of the value of GDP in 2000. The cyclical movement of productivity growth calculated by the LMS and by the BLS are nearly identical.

Importance of labor force participation

The fifth slider reflects the overall labor for participation rate (LFPR). The relationship between job growth and the unemployment rate depends on population growth and the LFPR. Changes in

demographics or behavior will change the willingness of the population to supply its labor (see several papers on this subject in the additional reading section below).
For example, as a greater share of the population approaches retirement age, the overall labor force participation rate will fall. As women dramatically started to enter the labor force in the 1970s and 1980s, the overall labor force participation rate increased.

Default assumptions

The LMS makes five assumptions. The population is assumed to grow at a rate equal to the average monthly population growth during the previous 12 months (excluding months that contain population adjustments from the U.S. Census Bureau). The CES/CPS multiplier (which translates the average monthly growth in payroll employment into the household concept of the unemployment rate, and vice versa) is set equal to the average of that ratio over the previous 12 months. Another tool, the Jobs Calculator, allows the user to explore the relationship between the UR and employment growth at different levels of population growth and for different values of the CES/CPS multiplier.

Starting values

The unemployment rate and the labor force participation rate are set equal to their current monthly value as reported by the BLS. The average change in monthly payroll employment is set equal to the value consistent with this UR and LFPR. Average annual productivity growth is constructed using the current level of GDP and payroll employment *[GDP*1/*E1*]. Average annual GDP growth reflects the growth that is consistent with the values reflected on the rest of the sliders.

Additional reading

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